

Supplementary Material: A Controlled Empirical Comparison of Classical and Quantum Kernel SVMs for Breast Cancer Classification

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Permanent Repository Tag: v1.0.0 (<https://github.com/SinaQP/SVM-Vs-QSVM>)

Primary Endpoint: Malignant Class F1 Score (pos_label=0)

Supplementary Section S1: Extended Sample-Size Scaling Evaluation

Models were trained on nested subsets $N_{\text{train}} \in \{50, 100, 200, 300, 455\}$ with preprocessing pipelines (`StandardScaler`, `PCA`, `MinMaxScaler`) independently refit on each subset and evaluated on fixed outer test partitions ($N_{\text{test}} = 114$). All learning curves use fixed classifier settings: $C = 1$ for every SVC, `gamma='scale'` for RBF, and `reps=1` with full entanglement for QSVC. They are separate from the inner-selected canonical comparison.

Table S1: Complete Sample-Size Scaling Results across 5 Outer Splits (Mean $\pm$ SD).

N_{train}	Model Architecture	Family	Repr.	Qubits	Test Accuracy	Precision	Recall	Malignant F1	ROC-AUC
50	Linear SVM	Classical	PCA 2	0	0.9333 $\pm$ 0.0354	0.8726 $\pm$ 0.0699	0.9667 $\pm$ 0.0271	0.9157 $\pm$ 0.0402	0.9874 $\pm$ 0.0087
50	RBF SVM	Classical	PCA 2	0	0.9263 $\pm$ 0.0282	0.9006 $\pm$ 0.0603	0.9048 $\pm$ 0.0532	0.9008 $\pm$ 0.0353	0.9839 $\pm$ 0.0072
50	QSVC (reps=1, full)	Quantum	PCA 2	2	0.8246 $\pm$ 0.0387	0.8242 $\pm$ 0.0957	0.6762 $\pm$ 0.0643	0.7398 $\pm$ 0.0531	0.8850 $\pm$ 0.0484
50	Linear SVM	Classical	PCA 4	0	0.9316 $\pm$ 0.0353	0.8857 $\pm$ 0.0893	0.9476 $\pm$ 0.0261	0.9128 $\pm$ 0.0395	0.9894 $\pm$ 0.0090
50	RBF SVM	Classical	PCA 4	0	0.9421 $\pm$ 0.0295	0.9197 $\pm$ 0.0607	0.9286 $\pm$ 0.0558	0.9222 $\pm$ 0.0385	0.9864 $\pm$ 0.0060
50	QSVC (reps=1, full)	Quantum	PCA 4	4	0.7263 $\pm$ 0.0716	0.6931 $\pm$ 0.1444	0.4095 $\pm$ 0.2146	0.4994 $\pm$ 0.2110	0.7636 $\pm$ 0.1082
100	Linear SVM	Classical	PCA 2	0	0.9474 $\pm$ 0.0175	0.9311 $\pm$ 0.0454	0.9286 $\pm$ 0.0412	0.9287 $\pm$ 0.0227	0.9884 $\pm$ 0.0067
100	RBF SVM	Classical	PCA 2	0	0.9246 $\pm$ 0.0237	0.9086 $\pm$ 0.0585	0.8905 $\pm$ 0.0686	0.8966 $\pm$ 0.0317	0.9849 $\pm$ 0.0092
100	QSVC (reps=1, full)	Quantum	PCA 2	2	0.8614 $\pm$ 0.0190	0.8679 $\pm$ 0.0647	0.7429 $\pm$ 0.0391	0.7983 $\pm$ 0.0207	0.9214 $\pm$ 0.0101
100	Linear SVM	Classical	PCA 4	0	0.9421 $\pm$ 0.0288	0.9246 $\pm$ 0.0675	0.9238 $\pm$ 0.0488	0.9222 $\pm$ 0.0355	0.9890 $\pm$ 0.0074
100	RBF SVM	Classical	PCA 4	0	0.9386 $\pm$ 0.0139	0.9256 $\pm$ 0.0380	0.9095 $\pm$ 0.0639	0.9155 $\pm$ 0.0217	0.9884 $\pm$ 0.0044
100	QSVC (reps=1, full)	Quantum	PCA 4	4	0.7947 $\pm$ 0.0332	0.8059 $\pm$ 0.0713	0.6048 $\pm$ 0.1722	0.6733 $\pm$ 0.0978	0.8628 $\pm$ 0.0326
200	Linear SVM	Classical	PCA 2	0	0.9421 $\pm$ 0.0159	0.9008 $\pm$ 0.0237	0.9476 $\pm$ 0.0310	0.9234 $\pm$ 0.0213	0.9867 $\pm$ 0.0079
200	RBF SVM	Classical	PCA 2	0	0.9316 $\pm$ 0.0157	0.9157 $\pm$ 0.0451	0.9000 $\pm$ 0.0391	0.9066 $\pm$ 0.0202	0.9864 $\pm$ 0.0045
200	QSVC (reps=1, full)	Quantum	PCA 2	2	0.8877 $\pm$ 0.0144	0.8821 $\pm$ 0.0521	0.8095 $\pm$ 0.0753	0.8406 $\pm$ 0.0259	0.9453 $\pm$ 0.0115
200	Linear SVM	Classical	PCA 4	0	0.9579 $\pm$ 0.0157	0.9432 $\pm$ 0.0262	0.9429 $\pm$ 0.0213	0.9429 $\pm$ 0.0210	0.9910 $\pm$ 0.0029
200	RBF SVM	Classical	PCA 4	0	0.9509 $\pm$ 0.0159	0.9311 $\pm$ 0.0382	0.9381 $\pm$ 0.0398	0.9337 $\pm$ 0.0216	0.9907 $\pm$ 0.0047
200	QSVC (reps=1, full)	Quantum	PCA 4	4	0.8579 $\pm$ 0.0287	0.8532 $\pm$ 0.0598	0.7476 $\pm$ 0.0686	0.7943 $\pm$ 0.0440	0.9298 $\pm$ 0.0148
300	Linear SVM	Classical	PCA 2	0	0.9439 $\pm$ 0.0268	0.9118 $\pm$ 0.0589	0.9429 $\pm$ 0.0361	0.9258 $\pm$ 0.0328	0.9892 $\pm$ 0.0038
300	RBF SVM	Classical	PCA 2	0	0.9333 $\pm$ 0.0133	0.9242 $\pm$ 0.0403	0.8952 $\pm$ 0.0494	0.9080 $\pm$ 0.0191	0.9864 $\pm$ 0.0047
300	QSVC (reps=1, full)	Quantum	PCA 2	2	0.9158 $\pm$ 0.0260	0.9193 $\pm$ 0.0448	0.8476 $\pm$ 0.0598	0.8807 $\pm$ 0.0388	0.9651 $\pm$ 0.0130
300	Linear SVM	Classical	PCA 4	0	0.9596 $\pm$ 0.0078	0.9441 $\pm$ 0.0240	0.9476 $\pm$ 0.0310	0.9453 $\pm$ 0.0112	0.9943 $\pm$ 0.0019
300	RBF SVM	Classical	PCA 4	0	0.9544 $\pm$ 0.0144	0.9493 $\pm$ 0.0444	0.9286 $\pm$ 0.0376	0.9377 $\pm$ 0.0188	0.9936 $\pm$ 0.0037
300	QSVC (reps=1, full)	Quantum	PCA 4	4	0.8877 $\pm$ 0.0259	0.8725 $\pm$ 0.0539	0.8190 $\pm$ 0.0598	0.8429 $\pm$ 0.0351	0.9489 $\pm$ 0.0156
455	Linear SVM	Classical	PCA 2	0	0.9474 $\pm$ 0.0152	0.9152 $\pm$ 0.0444	0.9476 $\pm$ 0.0261	0.9302 $\pm$ 0.0182	0.9897 $\pm$ 0.0026
455	RBF SVM	Classical	PCA 2	0	0.9439 $\pm$ 0.0159	0.9339 $\pm$ 0.0411	0.9143 $\pm$ 0.0271	0.9233 $\pm$ 0.0202	0.9864 $\pm$ 0.0040
455	QSVC (reps=1, full)	Quantum	PCA 2	2	0.9123 $\pm$ 0.0224	0.9313 $\pm$ 0.0269	0.8238 $\pm$ 0.0706	0.8726 $\pm$ 0.0376	0.9697 $\pm$ 0.0141
455	Linear SVM	Classical	PCA 4	0	0.9632 $\pm$ 0.0157	0.9493 $\pm$ 0.0344	0.9524 $\pm$ 0.0376	0.9501 $\pm$ 0.0211	0.9952 $\pm$ 0.0030
455	RBF SVM	Classical	PCA 4	0	0.9579 $\pm$ 0.0200	0.9448 $\pm$ 0.0436	0.9429 $\pm$ 0.0361	0.9430 $\pm$ 0.0259	0.9944 $\pm$ 0.0035
455	QSVC (reps=1, full)	Quantum	PCA 4	4	0.9053 $\pm$ 0.0423	0.8748 $\pm$ 0.0767	0.8762 $\pm$ 0.0832	0.8721 $\pm$ 0.0556	0.9581 $\pm$ 0.0227

Supplementary Section S2: Computational Complexity and Execution Profiling

The reported `total_runtime` includes preprocessing, classifier fitting, and prediction; QSVC additionally includes exact statevector generation and both training and test Gram products. Inner-search and kernel-diagnostic time are excluded.

Table S2: Execution Time, Computational Complexity, and Storage Footprint.

Pipeline Stage / Architecture	Platform	Qubits	Mean Runtime (s)	Rel. Cost	Time Complexity	Memory Complexity
Linear SVM (PCA 2)	CPU LibSVM	0	0.0048 $\pm$ 0.0007	1.0 $\times$ (Ref)	$\mathcal{O}(dN_{tr}^2) - \mathcal{O}(dN_{tr}^3)$, data dependent	Data/cache dependent
RBF SVM (PCA 2)	CPU LibSVM	0	0.0072 $\pm$ 0.0024	1.5 $\times$	Same SVC bounds; pairwise RBF work	Data/cache dependent
Linear SVM (PCA 4)	CPU LibSVM	0	0.0083 $\pm$ 0.0061	1.7 $\times$	$\mathcal{O}(dN_{tr}^2) - \mathcal{O}(dN_{tr}^3)$, data dependent	Data/cache dependent
RBF SVM (PCA 4)	CPU LibSVM	0	0.0071 $\pm$ 0.0019	1.5 $\times$	Same SVC bounds; pairwise RBF work	Data/cache dependent
QSVC (2Q Statevector)	Vectorized CPU	2	0.2257 $\pm$ 0.0232	$\sim 47\times$	$\mathcal{O}((N_{tr}^2 + N_{te}N_{tr})2^q)$ plus SVC	Train+test kernels: 2,071,160 B
QSVC (4Q Statevector)	Vectorized CPU	4	0.6599 $\pm$ 0.0863	$\sim 80\times$	Same form with larger q plus SVC	Train+test kernels: 2,071,160 B
<i>Historical 2Q ComputeUncompute</i>	Local circuit-pair sampler	2	120.5 $\pm$ 15.2	$\sim 25,000\times$	$\mathcal{O}(N_{tr}^2 + N_{te}N_{tr})$ pairs	Train+test Gram arrays
<i>Historical 4Q ComputeUncompute</i>	Local circuit-pair sampler	4	455.0 $\pm$ 40.0	$\sim 55,000\times$	$\mathcal{O}(N_{tr}^2 + N_{te}N_{tr})$ pairs	Train+test Gram arrays

The exact statevector and historical `ComputeUncompute` measurements are separate CPU-simulation regimes. Neither is a physical-QPU timing, and no fixed hardware shot budget is inferred.

Supplementary Section S3: Supplementary Diagnostic Figures

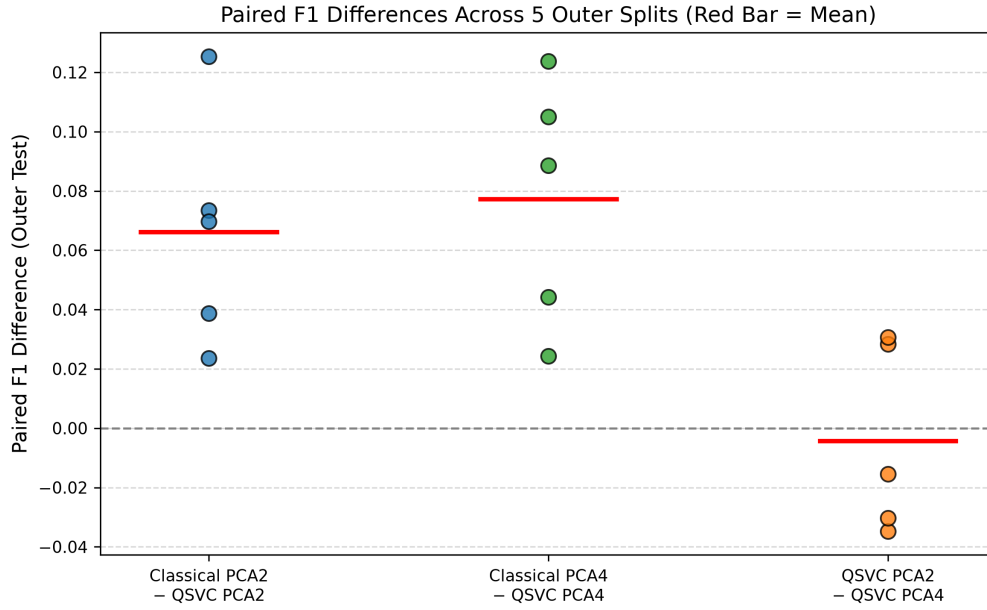


Figure S1: **Paired Split-Level F1 Differences.** Distribution of paired differences $\Delta = F1_{\text{Classical}} - F1_{\text{QSVc}}$ across five outer test splits. Classical models achieve higher malignant F1 scores on 5/5 splits in both PCA 2 and PCA 4 representations.

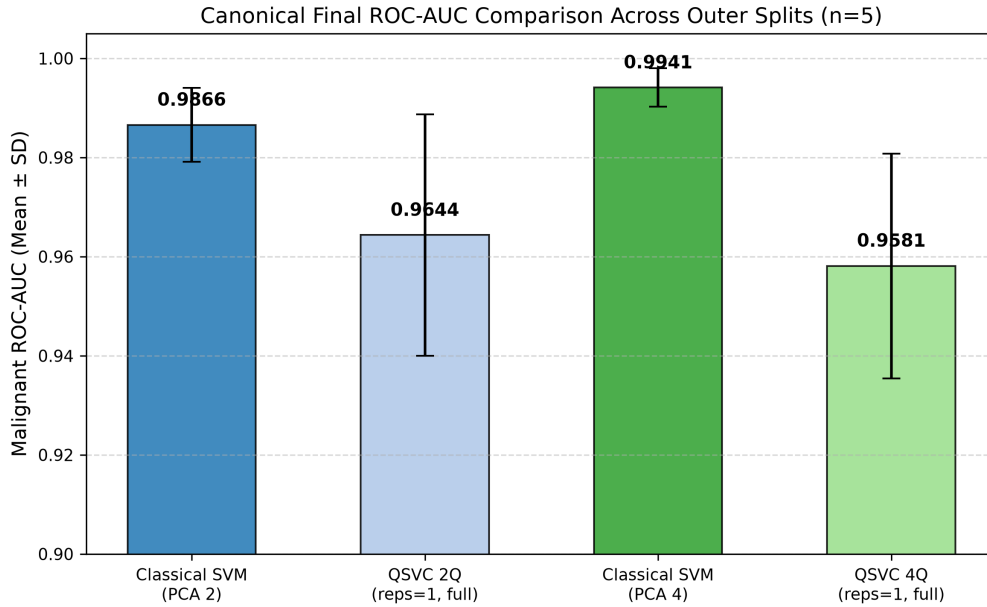


Figure S2: **Outer-Test ROC-AUC Comparison.** Mean receiver operating characteristic area under curve across five outer splits comparing classical and quantum models.